Notes on Computational Fluid Dynamics: General Principles

5.10 Pressure-velocity coupling

The previous section combined equations for $\mathbf{u}$ and p, governing momentum and mass conservation, in a sequential solution.

The algorithms used to couple these equations, in a manner which is convergent, uses the following notation to describe terms in the momentum equation, *e.g.* Eq. (2.67), *excluding* ∇p

$$A\mathbf{u} - \mathbf{H}(\mathbf{u}) \equiv \frac{\partial \mathbf{u}}{\partial t} + \nabla \bullet (\mathbf{u}\mathbf{u}) - \nabla \bullet (\nu \nabla \mathbf{u}) - \frac{\rho(T,p)}{\rho_0}\mathbf{g}, \tag{5.16}$$

where: $$\frac{\partial \mathbf{u}}{\partial t} + \nabla \bullet (\mathbf{u}\mathbf{u}) - \nabla \bullet (\nu \nabla \mathbf{u}) = -\nabla p + \frac{\rho(T,p)}{\rho_0}\mathbf{g} \tag{2.67}$$

- $A\mathbf{u}$ is a linear term in $\mathbf{u}$;
- $-\mathbf{H}(\mathbf{u})$ is a function of $\mathbf{u}$ and other sources.

Momentum corrector

A *momentum corrector equation* is formed by expressing the momentum equation, *e.g.* Eq. (2.67) in terms of the notation of Eq. (5.16), and rearranging to give

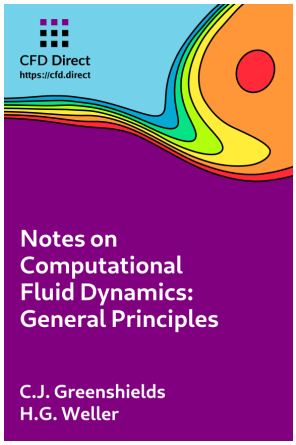
latest $\mathbf{u}$ and p

$$\mathbf{u} := \frac{\mathbf{H}(\mathbf{u})}{A} - \frac{1}{A}\nabla p. \tag{5.17}$$

先计算u再计算p，再使用最新的p去修正u

The equation provides an update to $\mathbf{u}$, based on current values of $\mathbf{u}$ and p substituted on the r.h.s. In other words, $\mathbf{H}(\mathbf{u})$ and ∇p are calculated explicitly.

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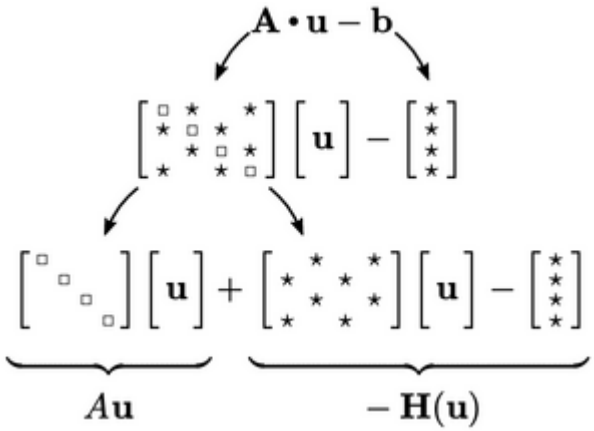


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In the algorithms described in the following sections, A is chosen to be the extracted diagonal coefficients $a_{i,i}\ i = 1, 2, \ldots, N$ of the matrix $\mathbf{A}$ corresponding to the momentum terms in Eq. (5.16).



The split between $A\mathbf{u}$ and $\mathbf{H}(\mathbf{u})$ is shown above. As a matrix with diagonal components only, A has one value per cell so can be represented as a scalar field. Setting A to be the diagonal coefficients is a natural choice for implicit treatment of $\mathbf{u}$ within the coupling algorithm.

Flux corrector

A *flux corrector equation* follows from Eq. (5.17) by interpolating $\mathbf{u}$ to cell faces and evaluating $\phi_\mathrm{f} = \mathbf{S}_\mathrm{f} \cdot \mathbf{u}_\mathrm{f}$ according to

$$\phi_\mathrm{f} := \mathbf{S}_\mathrm{f} \cdot \left(\frac{\mathbf{H}(\mathbf{u})}{A}\right)_\mathrm{f} - \left(\frac{|\mathbf{S}_\mathrm{f}|}{A}\right)_\mathrm{f} \nabla_\mathbf{n} p_\mathrm{f}. \tag{5.18}$$

Pressure equation

A *pressure equation* is then created by substituting fluxes ϕ_f from Eq. (5.18) into the mass conservation Eq. (2.46) in discrete form $\nabla \cdot \mathbf{u} \approx \sum_\mathrm{f} \phi_\mathrm{f} = 0$. The resulting expression is equivalent to a discretised pressure equation, with coefficients containing A and $\mathbf{H}(\mathbf{u})$,

$$\nabla \cdot \frac{1}{A} \nabla p = \nabla \cdot \left[\frac{\mathbf{H}(\mathbf{u})}{A}\right]. \tag{5.19}$$

Chris Greenshields Henry Weller 11th April 2022 Notes on CFD

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